

Multi-label subject classification in institutional repositories: comparing abstracts, keywords, and full text with SBERT, LaBSE, TF-IDF, and BM25

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Abstract

The volume of items in institutional repositories grows year after year, increasing the workload of cataloging staff. In this presentation, we introduce a reproducible workflow to support catalogers in assigning the “subject” metadata field through automatic multi-label classification in the SEDICI repository (Universidad Nacional de La Plata).

The pipeline integrates three components: (1) mapping bitstreams, CSV files, and handles to link full text exported from the DSpace Assetstore to repository records; (2) staged coverage measurement to estimate how many items are effectively “classifiable,” i.e., have available full text; and (3) stratified sampling to work with feasible subsets (e.g., 20,000 items) while preserving the subject distribution. Using these steps, we built a final dataset of 19,995 items with 37 subjects and produced a stratified train/validation/test split with no label loss across partitions.

We evaluated all possible input configurations: abstracts, keywords, and full text. For text representation, we compared TF-IDF, BM25, and two embedding-based alternatives (SBERT and LaBSE). Across all scenarios, BM25 achieved the best results with One-vs-Rest classifiers; the overall best configuration combined abstract + keywords + full text with BM25 + Stochastic Gradient Descent (SGD), reaching 70% effectiveness

¹ The authors’ contributions are declared following the TaDiRAH taxonomy (Taxonomy of Digital Research Activities in the Humanities). Carlos Javier Nusch was responsible for code development, natural language processing, data analysis, and manuscript writing, including tasks such as Programming, Modeling, Encoding, Analyzing, Cleaning, Evaluating, and Enriching. Leticia Cagnina, Ariel Lira, Gonzalo Villarreal, Leandro Antonelli, Marcelo Errecalde, and Marisa De Giusti contributed to content review and revision (Reviewing, Editing). Rafael Bértoli, Manuel Rojas, and María Marta Vila carried out data extraction, export, and preparation from the institutional DSpace platform (Extracting, Collecting, Preparing), enabling the processing and analysis pipeline to begin.

(macro $F1 = 0.708$; micro $F1 = 0.756$). BM25 also yielded strong results in other combinations. By comparison, the best TF-IDF setup was keywords + full text with LinearSVC (macro $F1 = 0.659$), while the best embedding-based configurations performed lower (SBERT: macro $F1 = 0.534$; LaBSE: macro $F1 = 0.509$).

Keywords

multi-label classification, subject assignment, subjects, subject headings, full text, TF-IDF, BM25, SBERT, LaBSE

Audience

Repository managers, metadata curation teams, and developers who require practical guidance to implement or evaluate automated subject assignment, including coverage measurement, sampling strategies, and reproducible evaluation. Due to space constraints, we do not provide a detailed explanation of the techniques used here; these will be discussed in depth in the accompanying presentation. For a concise introduction to the underlying concepts and their rationale, see the article and presentation presented at BIREDIAL (Nusch et al., 2025).

Time Zone

America/Argentina/Buenos_Aires (UTC−03:00)

Proposal

Motivation and objective

The sustained growth of scholarly output and the expansion of digital collections are generating an increasingly demanding volume of items for repository staff. Subject assignment is generally performed manually, and this metadata is a critical input for information retrieval and for organizing items within a repository. Our goal is to provide a reproducible and adaptable pipeline that helps prioritize and suggest subjects—without replacing the cataloger’s final decision—and that is understandable and reusable by repository teams. Ultimately, we aim to deliver a reliable methodology that can be integrated into a repository and used to suggest metadata to administrators, helping streamline their work.

What we did

To build the required dataset, we performed three exports from DSpace: (1) an export of all item-level metadata in the repository; (2) an export mapping each item’s handle to its corresponding internal identifiers in the DSpace Assetstore; and (3) an export of all full-text files stored in the Assetstore.

We then implemented a workflow designed to operate on large collections without requiring special infrastructure:

1. Text–record linking: we mapped bitstreams to handles using a correspondence CSV and converted the exported full-text files into TXT files.
2. Staged coverage measurement: we measured how many metadata CSV records had a handle, how many had at least one associated TXT, and finally how many had a non-empty full text.
3. Feasible sampling: we selected a target subset (20,000 handles) using multi-label stratified sampling (i.e., ensuring that all subject labels were represented).

4. *Dataset construction*: we grouped items by handle, combining their abstracts, keywords, and full text (truncated to 100,000 characters per item), and filtered to the 37 most frequent subjects (minimum support: 5 examples²).
5. *Stratified splitting*: we produced a train/validation/test split with automatic retries to avoid labels missing from any partition.
6. Finally, we evaluated how different combinations of input features and text representations affected the performance of multiple classification algorithms.

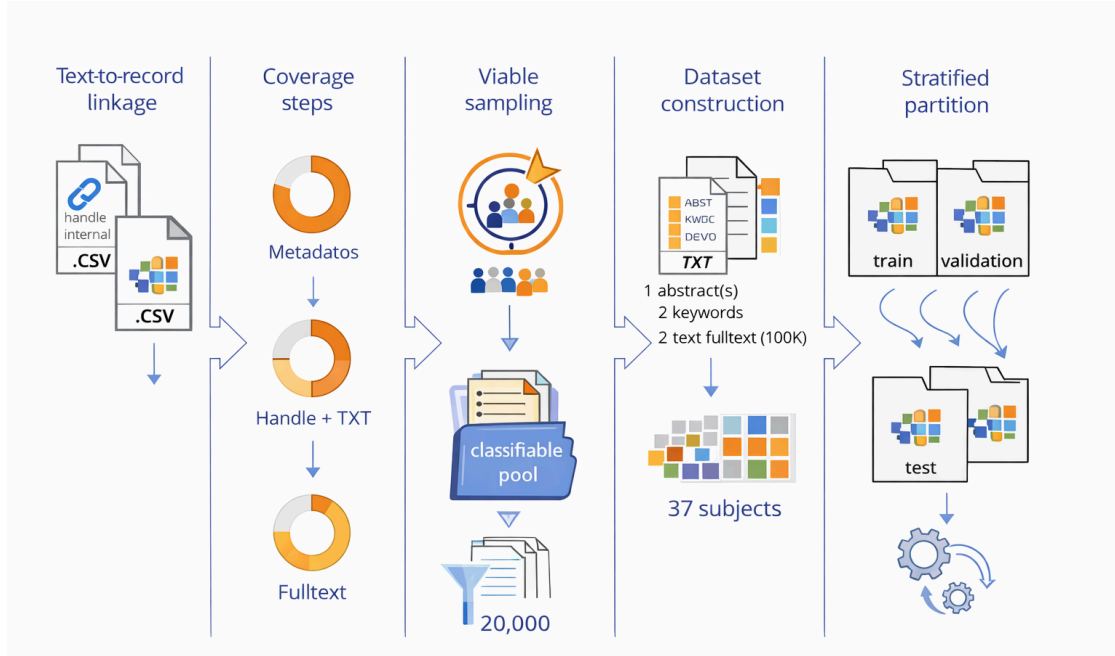


Figure 1. Dataset Preparation Workflow. Author’s own work.

Evaluation Design

We evaluated how different combinations of input features (abstracts, keywords, full text), represented with traditional techniques—specifically Term Frequency–Inverse Document Frequency (TF-IDF) and Okapi BM25 (Best Matching 25) (Robertson et al., 1994; Salton & McGill, 1986)—as well as more recent approaches based on large language model–based embeddings (SBERT and LaBSE) (Feng et al., 2022; Reimers & Gurevych, 2019), affected the performance of several classification algorithms: Logistic Regression (LogReg) (Jerome Hastie, 2003; Pedregosa et al., 2011), Linear Support Vector Machines (LinearSVC) (Fan et al., 2008), and Stochastic Gradient Descent (SGD) (Bottou, 2010). We report Accuracy, micro-F1, and macro-F1 on the validation set.

² For a detailed explanation of the coverage analysis performed and the resulting set of 37 labels, see the article and presentation presented at BIREDIAL (Nusch et al., 2025).

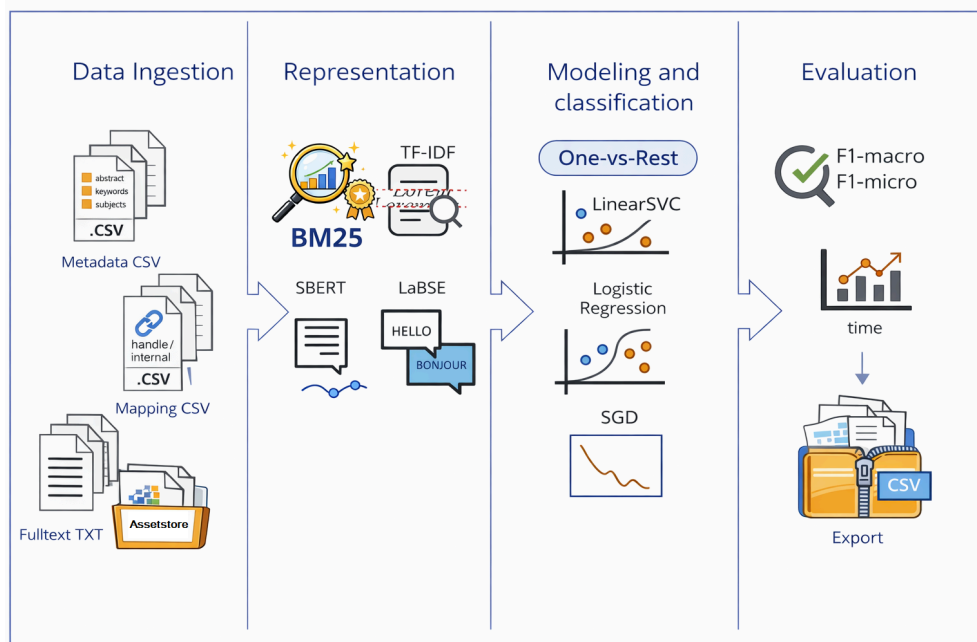


Figure 2. Evaluation Workflow. Author's own work.

Main Results

BM25 consistently outperformed the other representations across all source combinations. The best overall configuration combined all three sources (abstract + keywords + full text) with BM25 + SGD (macro F1 = 0.708; micro F1 = 0.756).

The best results per source combination (Top-1 / Top-2) were:

- abstract: BM25+SGD (0.466 / 0.522); BM25+LinearSVC (0.432 / 0.506)
- keywords: BM25+LinearSVC (0.417 / 0.471); BM25+SGD (0.405 / 0.475)
- full text: BM25+SGD (0.691 / 0.745); BM25+LinearSVC (0.679 / 0.736)
- abstract + keywords: BM25+SGD (0.501 / 0.561); BM25+LinearSVC (0.484 / 0.557)
- abstract + full text: BM25+SGD (0.699 / 0.748); BM25+LinearSVC (0.682 / 0.739)
- keywords + full text: BM25+SGD (0.699 / 0.747); BM25+LinearSVC (0.685 / 0.740)
- abstract + keywords + full text: BM25+SGD (0.708 / 0.756); BM25+LinearSVC (0.689 / 0.746)

As a reference point, the best TF-IDF result was keywords + full text + LinearSVC (macro F1 = 0.659; micro F1 = 0.727). Embedding-based alternatives achieved their best values with full text or keywords + full text, but remained below BM25 (SBERT: 0.534; LaBSE: 0.509). For a more detailed exploration of all results and evaluations, see Table 1 in the Annex.

Why it matters for managers and developers

Our contribution is not only “a model,” but an operational method:

1. It enables teams to estimate—using simple metrics—what proportion of a collection is realistically addressable with automatic classification (full-text coverage).
2. It reduces effort by avoiding full-text processing when it is not necessary, while still preserving the subject distribution through multi-label stratified sampling.
3. It provides comparative evidence to guide implementation decisions: when metadata alone is sufficient, when adding full text is beneficial, and which text representation is most effective.
4. It outputs reusable artifacts (mapping CSV, sampled subset, data splits, and results) that are ready to be integrated into pipelines and dashboards.

Reproducibility and transferability

We will provide an executable notebook that produces run-versioned outputs. The staged design makes it straightforward to adapt the workflow to other repositories with similar structures, requiring only the replacement of the mapping and metadata CSVs and the handle-extraction rules.

Limitations and next step

This experiment focused on an operationally feasible, multi-label stratified subset (20,000 items) and on 37 high-frequency subject labels. This enables a reproducible evaluation of the relative contribution of abstracts, keywords, and full text, but it leaves out the long tail of less frequent labels. As next steps, we plan to:

1. Expand coverage to lower-support subjects and study strategies to mitigate imbalance (e.g., class weighting and other reweighting schemes).
2. Improve data quality in abstracts and other textual fields (normalization, noise and redundancy removal, and—when beneficial—sentence segmentation to support downstream representations).
3. Optimize classifiers and training settings (e.g., iteration limits, convergence/early-stopping criteria, and systematic hyperparameter search), while preserving the workflow’s reproducibility.
4. Deepen representation experiments: beyond TF-IDF/BM25, evaluate alternative word and character *n*-gram configurations (to capture morphological patterns in academic language) and, for embeddings (SBERT/LaBSE), compare pooling/aggregation strategies.

From a practical adoption perspective, we will evaluate confidence thresholds to provide actionable suggestions to catalogers and explore integration as a support component within ingestion/curation workflows. Finally, to improve interoperability, a future line of work is to map or re-label local subjects to external controlled vocabularies (e.g., UNESCO, OECD Fields of Science, JEL), which requires defining mapping criteria and a transition plan from the local scheme.

References

- Bottou, L. (2010). Large-Scale Machine Learning with Stochastic Gradient Descent. En Y. Lechevallier & G. Saporta (Eds.), *Proceedings of COMPSTAT'2010* (pp. 177-186). Physica-Verlag HD. https://doi.org/10.1007/978-3-7908-2604-3_16
- Fan, R.-E., Chang, K.-W., Hsieh, C.-J., Wang, X.-R., & Lin, C.-J. (2008). LIBLINEAR: A Library for Large Linear Classification. *J. Mach. Learn. Res.*, 9, 1871-1874.
- Feng, F., Yang, Y., Cer, D., Arivazhagan, N., & Wang, W. (2022). *Language-agnostic BERT Sentence Embedding* (No. arXiv:2007.01852). arXiv. <https://doi.org/10.48550/arXiv.2007.01852>
- Jerome Hastie, T. T. (2003). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*.
- Nusch, C. J., Cagnina, L. C., Pelocche, S. B., Villarreal, G. L., Lira, A. J., Antonelli, R. L., Folegatto, L. E., Errecalde, M. L., & De Giusti, M. R. (2025). *Clasificación automática de materias en repositorios institucionales mediante aprendizaje supervisado y representaciones vectoriales multilingües: Un estudio de caso en SEDICI*. XIV Conferencia Internacional Biredial-ISTEC (Brasil, 8 al 10 de octubre de 2025). <http://sedici.unlp.edu.ar/handle/10915/185805>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine Learning in Python. *J. Mach. Learn. Res.*, 12(null), 2825-2830.
- Reimers, N., & Gurevych, I. (2019). *Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks* (No. arXiv:1908.10084). arXiv. <https://doi.org/10.48550/arXiv.1908.10084>
- Robertson, S., Walker, S., Jones, S., Hancock-Beaulieu, M., & Gatford, M. (1994). *Okapi at TREC-3*. Text Retrieval Conference.

Anexus

Table 1. Validation results by combination of textual sources and representation (TF-IDF, BM25, SBERT, LaBSE) using One-vs-Rest classifiers. Author's own work.

feature_set	combine_mode	repr	clf	acc	f1_micro	f1_macro	time_sec
all3	concat_prefixed	bm25	SGD	0.5973333333	0.7561692127	0.7080456761	76.58238006
abs+ft	concat_prefixed	bm25	SGD	0.591	0.7481074662	0.6992215331	66.53072596
kw+ft	concat_prefixed	bm25	SGD	0.5886666667	0.7473435655	0.6990508574	78.5358212
fulltext	concat_prefixed	bm25	SGD	0.5883333333	0.7449476324	0.6912620192	69.15497756
all3	concat_prefixed	bm25	LinearSVC	0.5813333333	0.7460639127	0.6890230032	218.555228
kw+ft	concat_prefixed	bm25	LinearSVC	0.5746666667	0.7402717727	0.6850176441	474.4729872
abs+ft	concat_prefixed	bm25	LinearSVC	0.5733333333	0.7385528989	0.6819087619	254.6651962
all3	concat_prefixed	bm25	LogReg	0.5733333333	0.7400253325	0.680006632	730.537163
fulltext	concat_prefixed	bm25	LinearSVC	0.5726666667	0.7358109159	0.6792677107	6142.419467
kw+ft	concat_prefixed	bm25	LogReg	0.5713333333	0.7390824531	0.678436025	788.7627733
fulltext	concat_prefixed	bm25	LogReg	0.5696666667	0.7357774968	0.6746947778	876.2804987
abs+ft	concat_prefixed	bm25	LogReg	0.5696666667	0.7351590945	0.6745324029	737.9700096
kw+ft	concat_prefixed	tfidf	LinearSVC	0.5566666667	0.7266256875	0.6591632294	105.6545999
fulltext	concat_prefixed	tfidf	LinearSVC	0.5516666667	0.7244732577	0.6583429092	108.3854799
all3	concat_prefixed	tfidf	LinearSVC	0.5483333333	0.7200520833	0.6521966362	96.34628534
abs+ft	concat_prefixed	tfidf	LinearSVC	0.544	0.7174302951	0.6495870719	105.385253
fulltext	concat_prefixed	sbert	LinearSVC	0.4053333333	0.5981050818	0.5336754426	29.51249337
kw+ft	concat_prefixed	sbert	LinearSVC	0.406	0.6010854817	0.532521343	25.78102851
kw+ft	concat_prefixed	labse	LinearSVC	0.3973333333	0.5901807024	0.5089646646	40.39710903
abs+kw	concat_prefixed	bm25	SGD	0.375	0.5605844053	0.5014437387	8.598210573
fulltext	concat_prefixed	labse	LinearSVC	0.394	0.5760663098	0.4979949833	45.72327924
abs+kw	concat_prefixed	bm25	LinearSVC	0.3736666667	0.5566545202	0.4843776096	608.1077173
abstract	concat_prefixed	bm25	SGD	0.3363333333	0.5220779221	0.4656825837	8.991740942
abs+kw	concat_prefixed	bm25	LogReg	0.3553333333	0.5417192287	0.4573682224	63.23642993
abs+kw	concat_prefixed	tfidf	LinearSVC	0.347	0.5318991098	0.4373901987	8.757164478
abstract	concat_prefixed	bm25	LinearSVC	0.318	0.5061371841	0.4324021396	1332.301824
all3	concat_prefixed	labse	LinearSVC	0.3406666667	0.5259728507	0.4186679111	44.38951755
keywords	concat_prefixed	bm25	LinearSVC	0.318	0.4714142427	0.4170092306	206.9555714
abs+ft	concat_prefixed	labse	LinearSVC	0.338	0.5214879018	0.4152545324	49.7968533
abs+kw	concat_prefixed	labse	LinearSVC	0.333	0.5153664303	0.406403589	49.88080287
keywords	concat_prefixed	bm25	SGD	0.3196666667	0.4746657284	0.4052926133	2.873643398

abstract	concat_prefixed	bm25	LogReg	0.306	0.4914457605	0.4048841685	69.85283732
keywords	concat_prefixed	tfidf	LinearSVC	0.3216666667	0.4857776136	0.4045049902	2.343902826
keywords	concat_prefixed	bm25	LogReg	0.3183333333	0.4720564297	0.3992641751	13.38760018
all3	concat_prefixed	sbert	LinearSVC	0.317	0.4914700544	0.3924540331	27.6888082
kw+ft	concat_prefixed	sbert	LogReg	0.3136666667	0.4985899605	0.3895296585	42.70876479
abs+kw	concat_prefixed	sbert	LinearSVC	0.3153333333	0.4868708972	0.3883354363	32.2016046
abs+ft	concat_prefixed	sbert	LinearSVC	0.316	0.4917016232	0.3872890121	33.31414676
abstract	concat_prefixed	labse	LinearSVC	0.313	0.4923989618	0.3855440774	53.71585536
abstract	concat_prefixed	sbert	LinearSVC	0.308	0.4816176471	0.3838284506	34.32724977
fulltext	concat_prefixed	sbert	LogReg	0.3086666667	0.4924372966	0.3774529245	45.05673695
abstract	concat_prefixed	tfidf	LinearSVC	0.293	0.4753050552	0.3773728376	8.068031788
kw+ft	concat_prefixed	labse	LogReg	0.3123333333	0.4968803176	0.3757763352	65.96144986
keywords	concat_prefixed	labse	LinearSVC	0.2933333333	0.4640585855	0.3623668835	71.33939219
keywords	concat_prefixed	sbert	LinearSVC	0.291	0.4591738713	0.3615529843	43.91433096
kw+ft	concat_prefixed	tfidf	LogReg	0.3033333333	0.4834066373	0.3510458493	367.7241294
fulltext	concat_prefixed	labse	LogReg	0.2926666667	0.4741081703	0.3500208492	71.14569378
fulltext	concat_prefixed	tfidf	LogReg	0.3006666667	0.479759519	0.3446920044	365.4790223
all3	concat_prefixed	tfidf	LogReg	0.2953333333	0.4735678392	0.3362182013	360.2791953
kw+ft	concat_prefixed	sbert	SGD	0.277	0.4605849312	0.3347471857	7.650916338
abs+ft	concat_prefixed	tfidf	LogReg	0.2933333333	0.4712041885	0.3323237539	360.5019724
fulltext	concat_prefixed	sbert	SGD	0.28	0.4521393035	0.3298567352	7.711553335
kw+ft	concat_prefixed	labse	SGD	0.2853333333	0.4597520173	0.3262210609	11.076262
all3	concat_prefixed	labse	LogReg	0.2756666667	0.4560928433	0.3226298789	65.11437702
abs+kw	concat_prefixed	labse	LogReg	0.2683333333	0.4471050049	0.3198966321	69.67728686
abs+ft	concat_prefixed	labse	LogReg	0.271	0.4482892691	0.3190564037	70.27020383
fulltext	concat_prefixed	labse	SGD	0.272	0.4447943318	0.3137690137	11.54437685
abs+kw	concat_prefixed	sbert	LogReg	0.2603333333	0.4270323213	0.3106163883	45.48185706
all3	concat_prefixed	sbert	LogReg	0.2583333333	0.4258317025	0.309450141	42.32326818
abs+ft	concat_prefixed	sbert	LogReg	0.2563333333	0.4222309505	0.3009245588	46.32483554
abstract	concat_prefixed	labse	LogReg	0.255	0.4282315623	0.2996942454	71.03631425
keywords	concat_prefixed	labse	LogReg	0.246	0.4081053698	0.2978192501	78.01565742
keywords	concat_prefixed	sbert	LogReg	0.2366666667	0.3973077708	0.2946367725	51.10671282
kw+ft	concat_prefixed	tfidf	SGD	0.26	0.4333402447	0.2937934747	46.77646089
abstract	concat_prefixed	sbert	LogReg	0.2493333333	0.4131121643	0.2918688463	46.53137088
fulltext	concat_prefixed	tfidf	SGD	0.2586666667	0.4317144043	0.2892870217	46.56326938
abs+kw	concat_prefixed	sbert	SGD	0.2426666667	0.4068	0.2791580249	8.236158133
abs+ft	concat_prefixed	tfidf	SGD	0.247	0.4163693599	0.2770894864	47.77137113
all3	concat_prefixed	tfidf	SGD	0.248	0.4183225267	0.2770889251	47.06952143
all3	concat_prefixed	labse	SGD	0.252	0.4230383303	0.2751587996	12.12574339

keywords	concat_prefixed	labse	SGD	0.2286666667	0.3875873407	0.2730562496	12.32633543
all3	concat_prefixed	sbert	SGD	0.2403333333	0.4041616647	0.2702210803	7.846075296
abs+ft	concat_prefixed	labse	SGD	0.2403333333	0.4129734085	0.2649474758	11.85975266
abs+ft	concat_prefixed	sbert	SGD	0.2326666667	0.3960515713	0.2640889528	8.422932625
keywords	concat_prefixed	sbert	SGD	0.2226666667	0.3760648244	0.2623200714	8.169344664
abs+kw	concat_prefixed	labse	SGD	0.2336666667	0.399507288	0.2622200181	11.59940577
abstract	concat_prefixed	sbert	SGD	0.2373333333	0.3926680244	0.2587675087	8.08254528
abstract	concat_prefixed	labse	SGD	0.2283333333	0.390615353	0.2516431837	11.64274359
keywords	concat_prefixed	tfidf	LogReg	0.163	0.2968118545	0.1948073893	6.404637814
keywords	concat_prefixed	tfidf	SGD	0.1316666667	0.2497674419	0.1494962656	1.409718513
abs+kw	concat_prefixed	tfidf	LogReg	0.1416666667	0.2617728532	0.1493280017	31.79569125
abstract	concat_prefixed	tfidf	LogReg	0.112	0.2054958184	0.101750491	31.79743123
abs+kw	concat_prefixed	tfidf	SGD	0.1053333333	0.2005772006	0.0994019785 2	5.524262428
abstract	concat_prefixed	tfidf	SGD	0.08	0.1496902107	0.0649261027 5	5.695212364